Fluid Automation

Laboratory report

Group 08

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Master Degree in Mechatronics Enginnering

2023|2024

Fluid Automation

Task 2

Bottle filling and sorting system



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Explanation

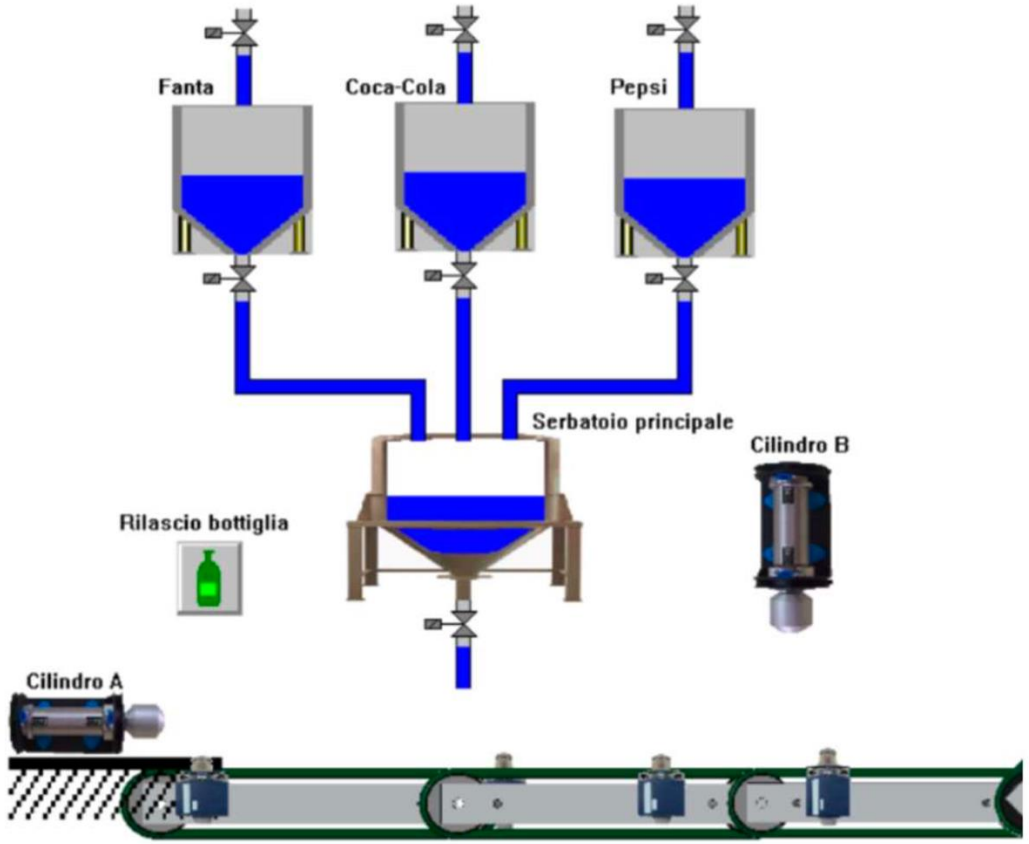
Bottles of 3 typologies have to be filled and sorted (Fanta, Coca-cola and Pepsi)

By means of proper **selectors** the operator can choose the bottle to be filled down the filling tank (**cylinder A** pushes the bottles and a conveyor moves them; conveyor must be stopped during the filling)

Cylinder B simulates bottle corking (tapping); conveyor must be stopped during corking

A **tank** at the end of the conveyor collects a predefined number of bottles of the same type

When the tank is full send the tank to a **sorting** system (a pneumatic cylinder will push the tank on the conveyor to start the motors)



Objectives

AUTOMATIC CONTROL

- Fluidsim
- CodeSys
- Ladder code
- Functional blocks
- Structured text

PNEUMATICS

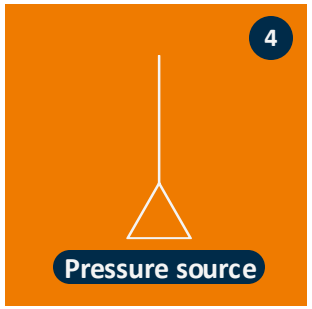
- Pneumatic cyliders
- Electrovalves

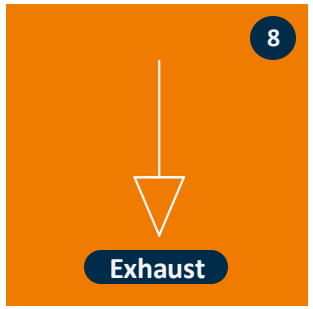
HMI INTERFACES

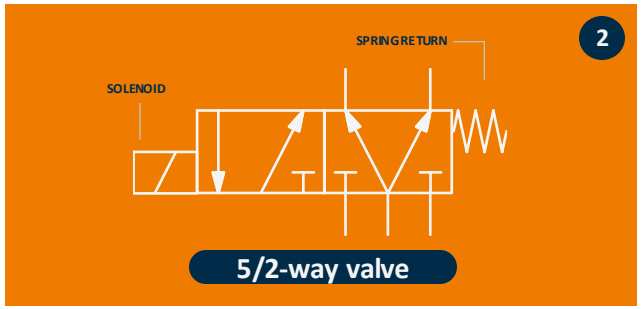
- Pushbuttons
- Timers
- Actuators
- Sensors
- Lights
- Alarms

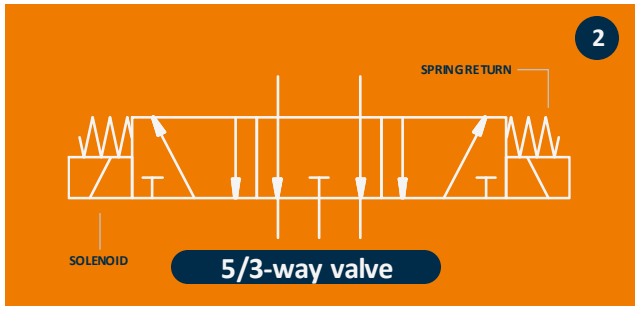


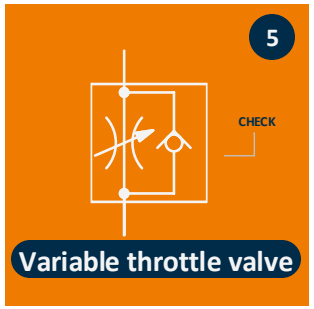
Components pneumatic

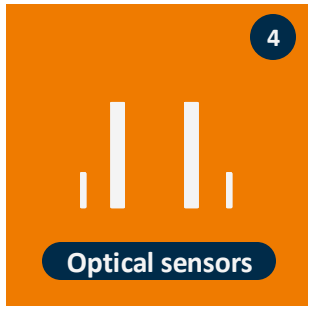


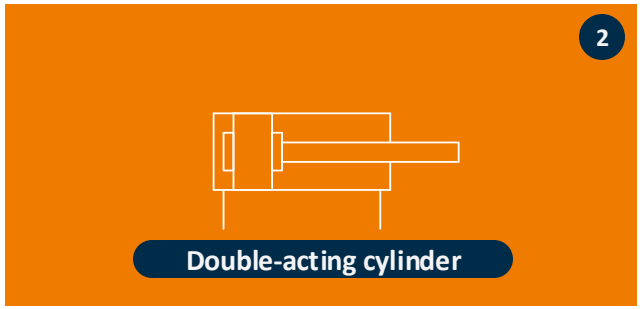


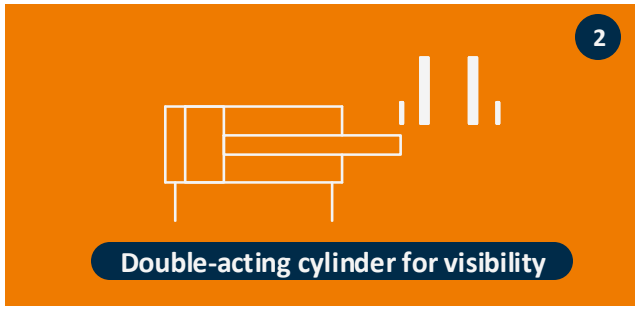




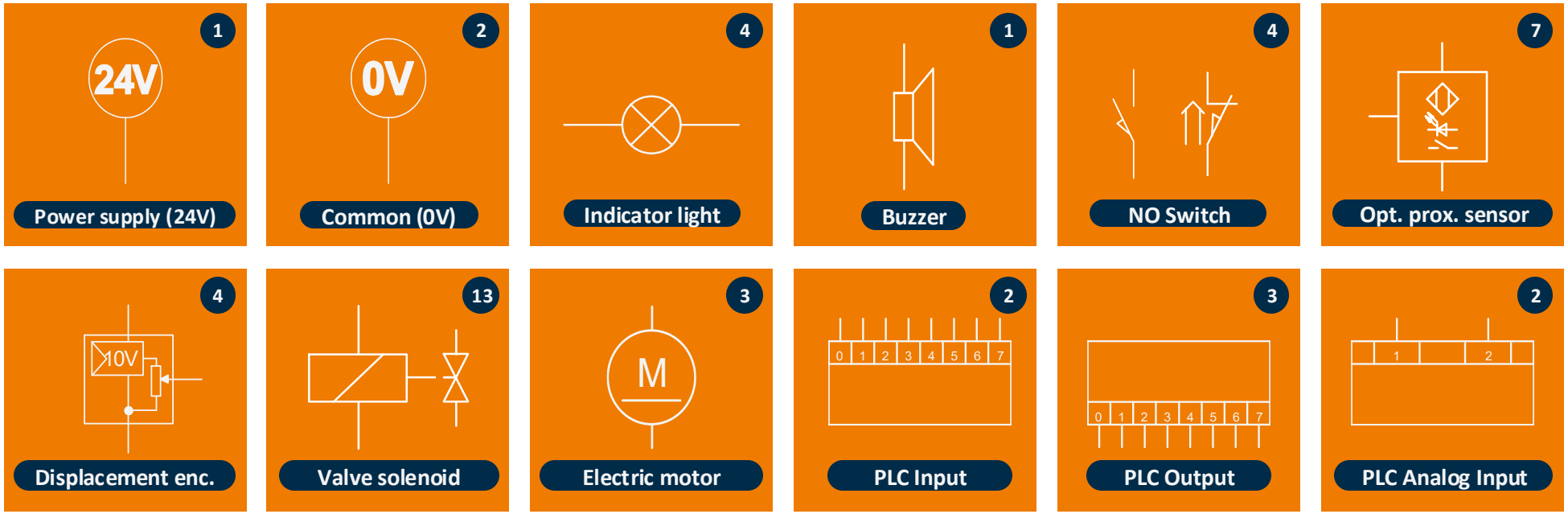




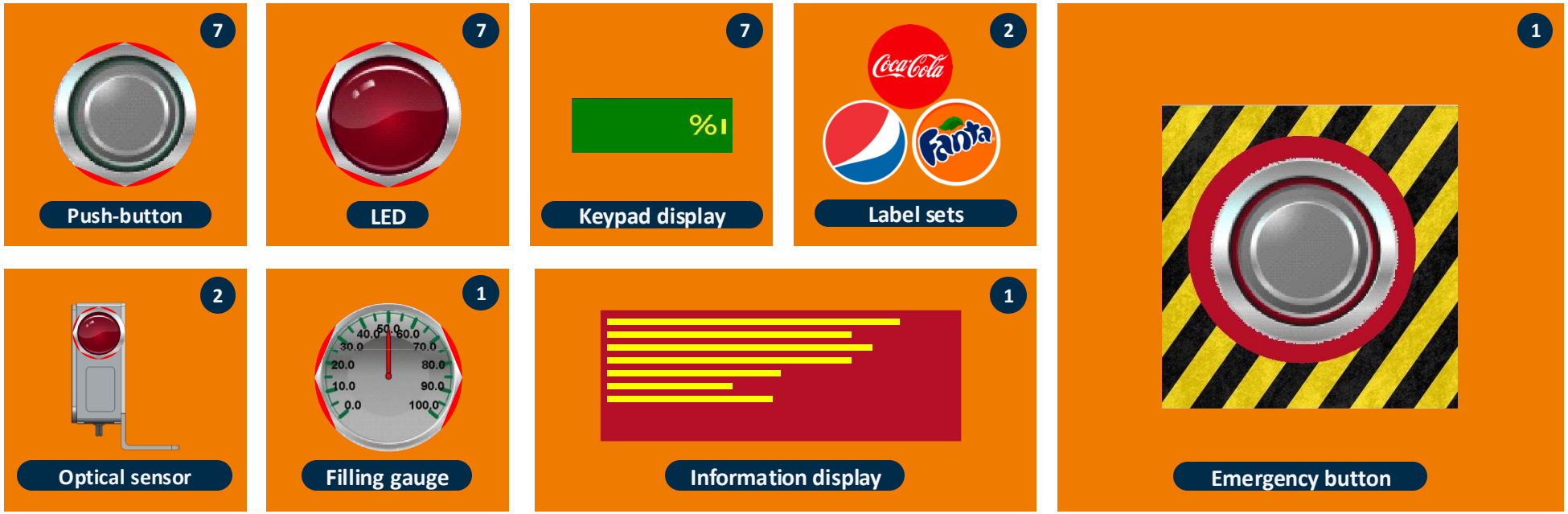




Components electrical



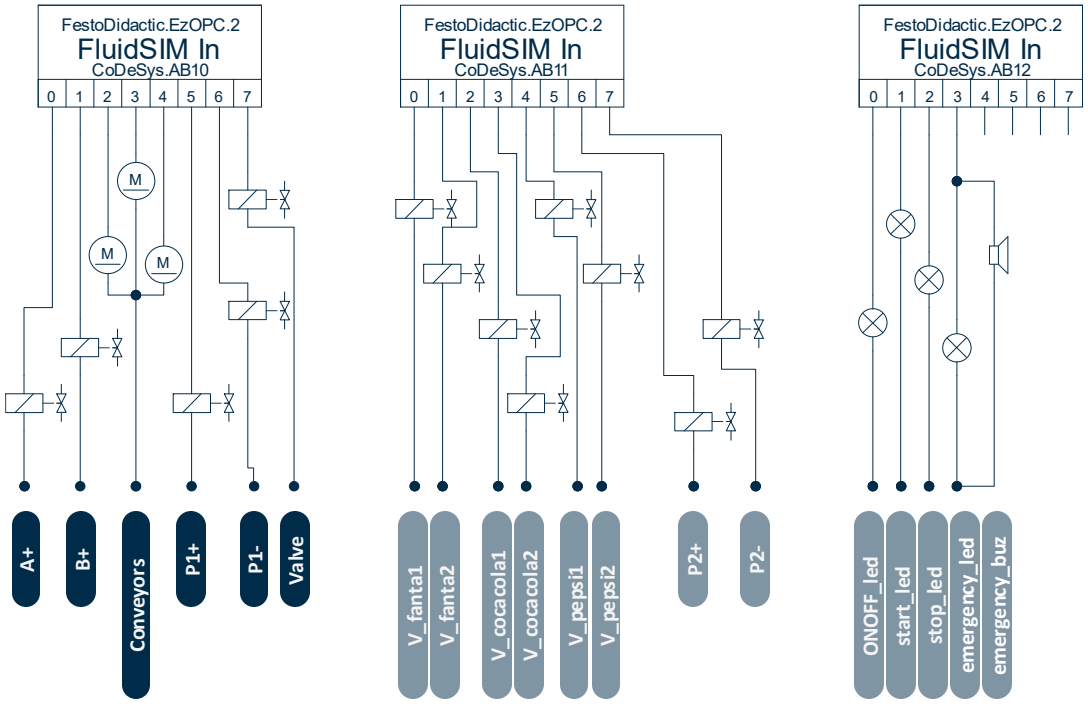
Components HMI



PLC

input connections

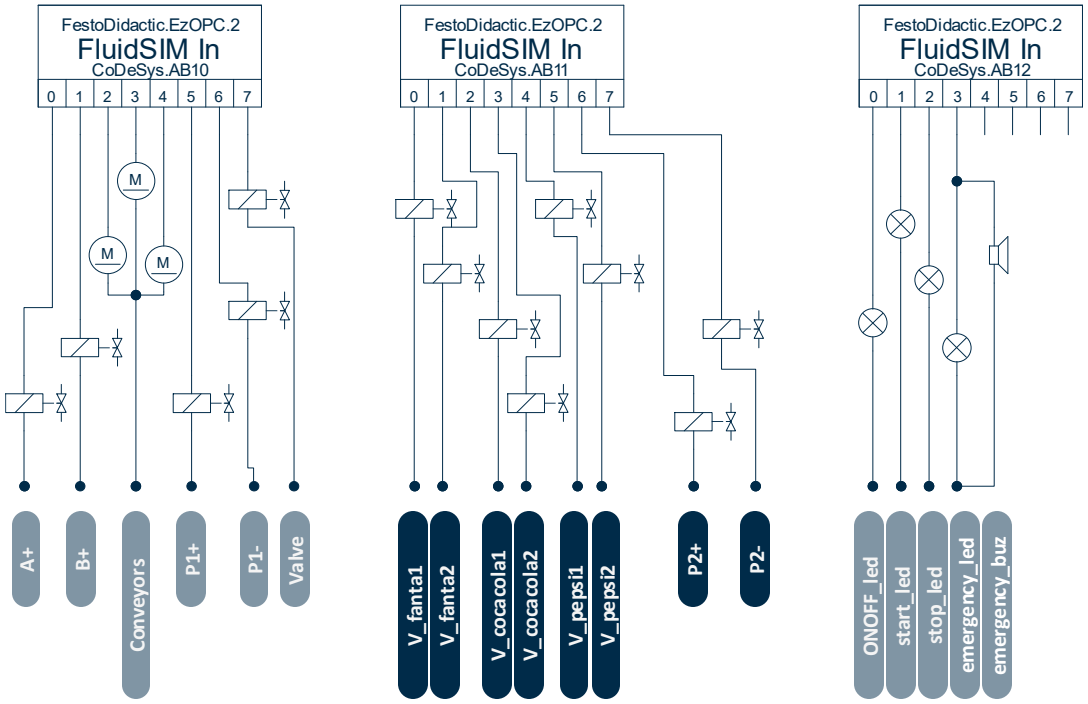
	component	functionality
AB10.IN0	solenoid	A+
AB10.IN1	solenoid	B+
AB10.IN2	motor	Conveyor 1
AB10.IN3	motor	Conveyor 2
AB10.IN4	motor	Conveyor 3
AB10.IN5	solenoid	P1+
AB10.IN6	solenoid	P1-
AB10.IN7	solenoid	Valve



PLC

input connections

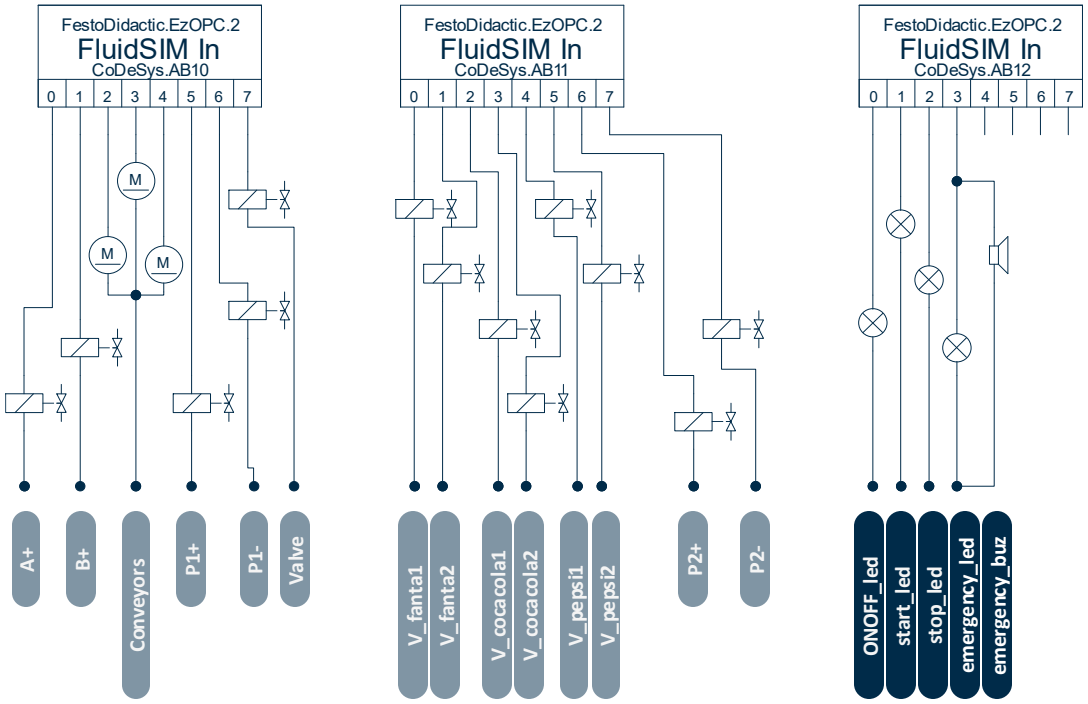
	component	functionality
AB11.IN0	solenoid	V_fanta1
AB11.IN1	solenoid	V_fanta2
AB11.IN2	solenoid	V_cocacola1
AB11.IN3	solenoid	V_cocacola2
AB11.IN4	solenoid	V_pepsi1
AB11.IN5	solenoid	V_pepsi2
AB11.IN6	solenoid	P2+
AB11.IN7	solenoid	P2-



PLC

input connections

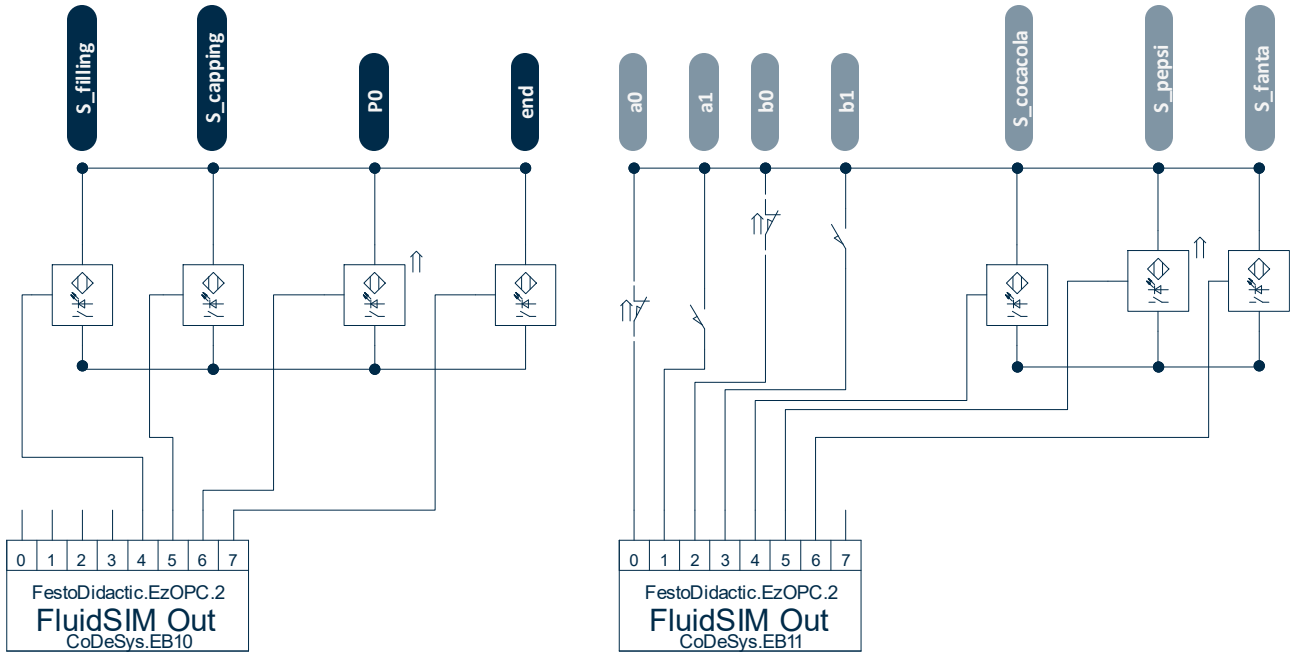
	component	functionality
AB12.IN0	led	ONOFF_led
AB12.IN1	led	start_led
AB12.IN2	led	stop_led
AB12.IN3	led	emergency_led
AB12.IN4	buzzer	emergency_buz



PLC

output connections

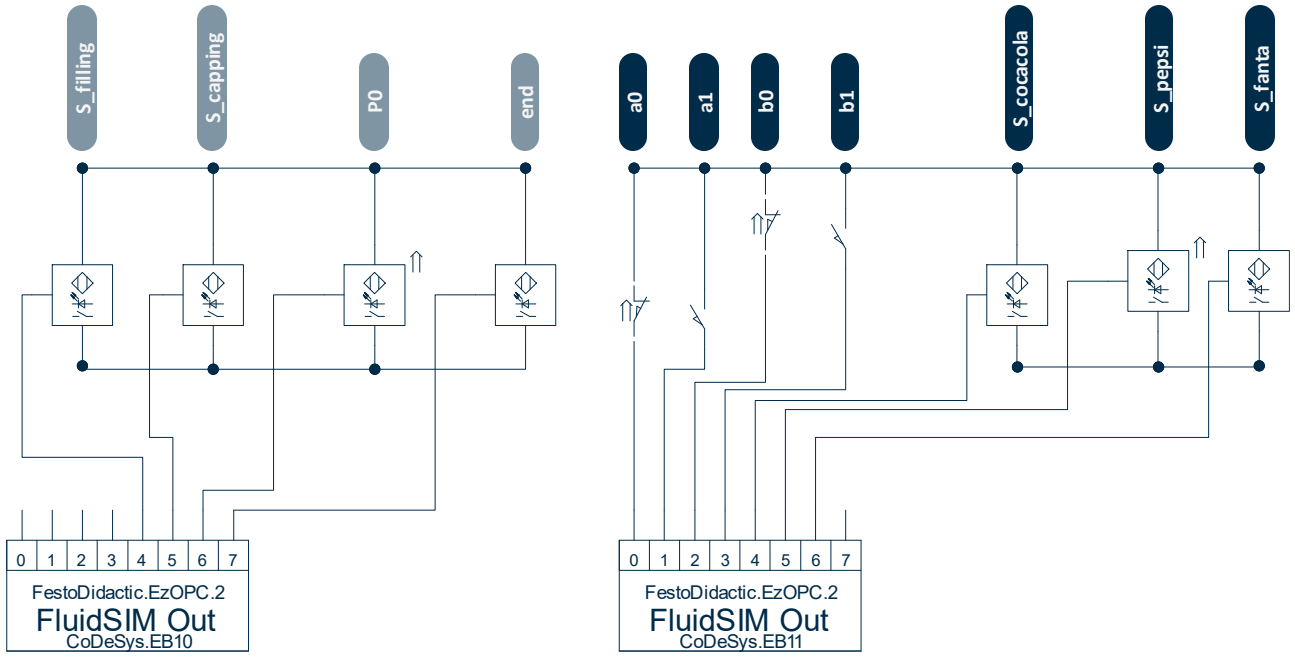
	component	functionality
EB10.OUT4	Optical proximity sensor	S_filling
EB10.OUT5	Optical proximity sensor	S_capping
EB10.OUT6	Optical proximity sensor	P0
EB10.OUT7	Optical proximity sensor	end



PLC

output connections

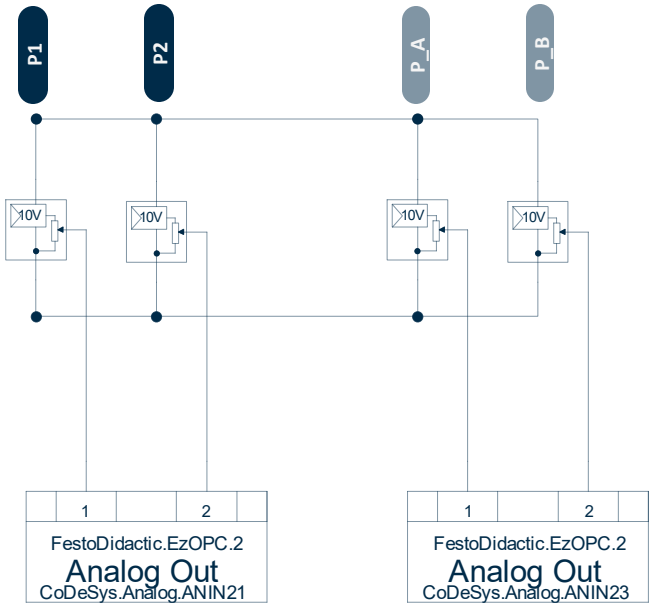
	component	functionality
EB11.OUT0	NO switch	a0
EB11.OUT1	NO switch	a1
EB11.OUT2	NO switch	b0
EB11.OUT3	NO switch	b1
EB11.OUT4	Optical proximity sensor	S_cocacola
EB11.OUT5	Optical proximity sensor	S_pepsi
EB11.OUT6	Optical proximity sensor	S_fanta



PLC

analog connections

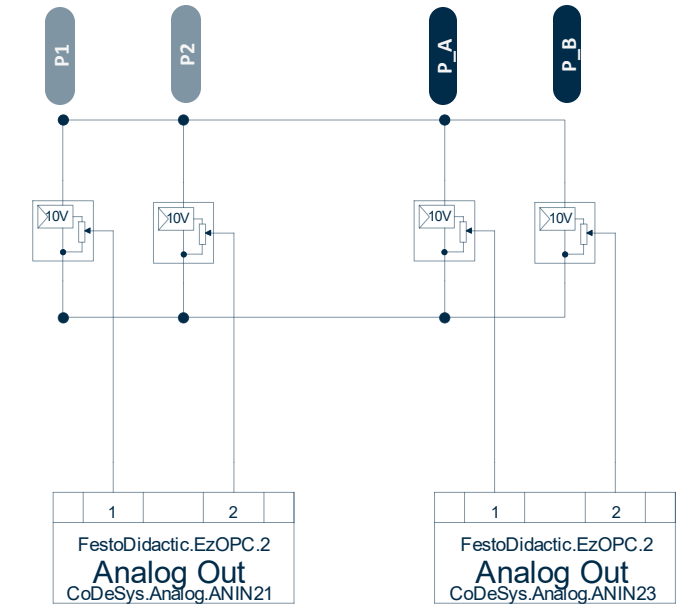
	component	functionality
ANIN21.OUT1	Displacement encoder	P1
ANIN21.OUT2	Displacement encoder	P2



PLC

analog connections

	component	functionality
ANIN23.OUT1	Displacement encoder	P_A
ANIN23.OUT2	Displacement encoder	P_B



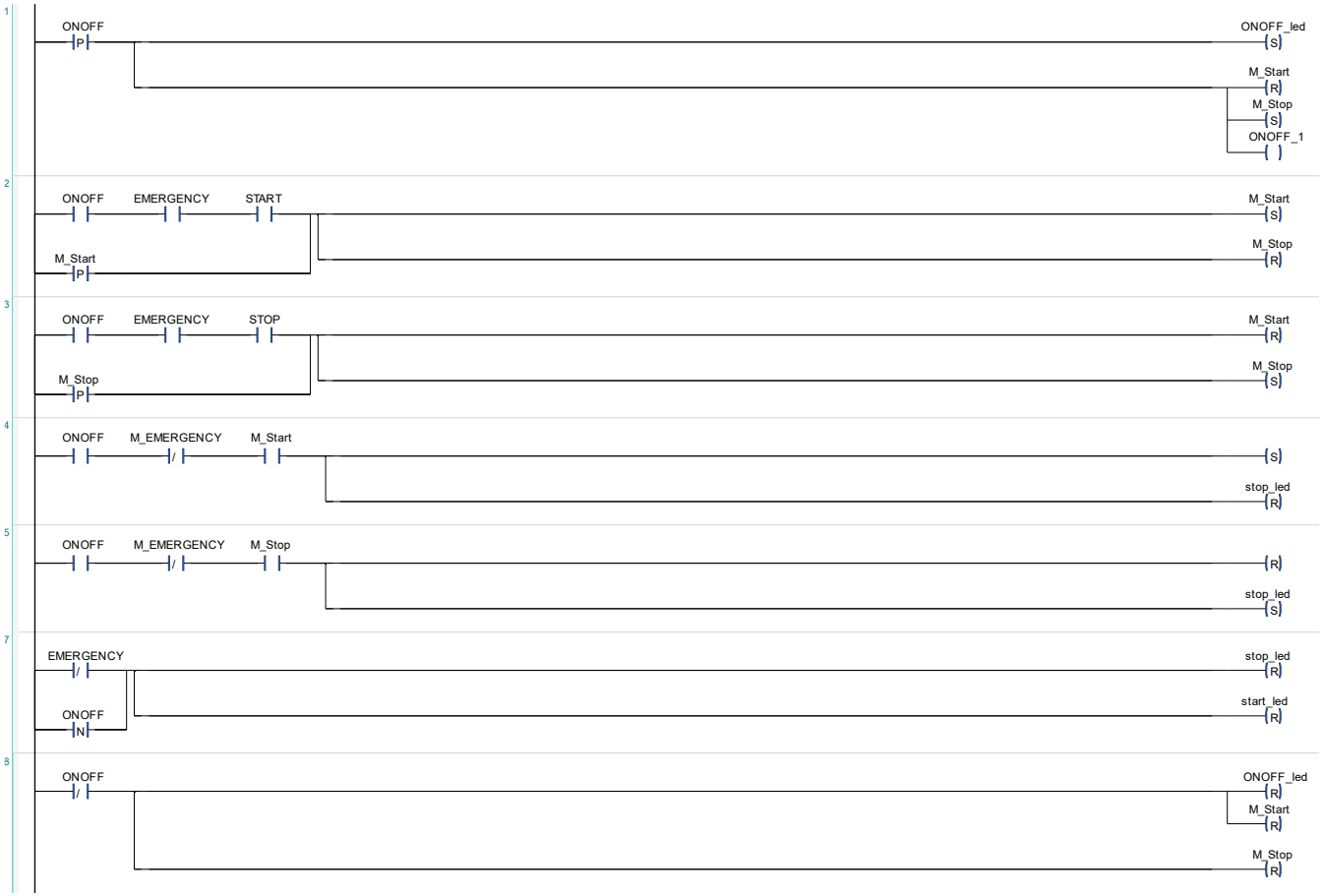
main

1 - 5 | 7 | 8 | LD

Memories for the **start** of the machine are controlled

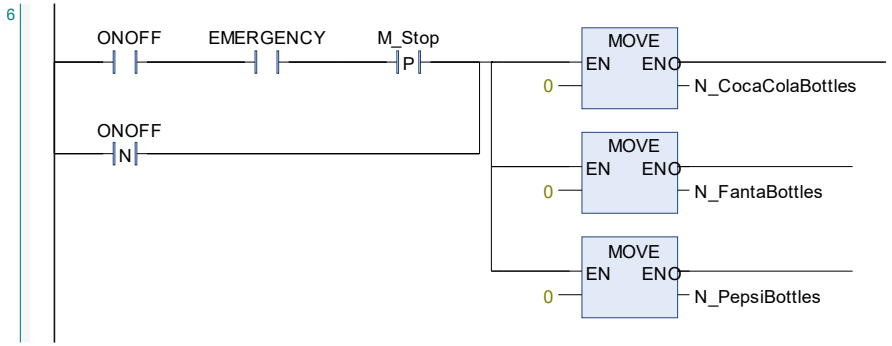
Memories for the **stop** of the machine are controlled

Lights are controlled by the memories



main 6 LD

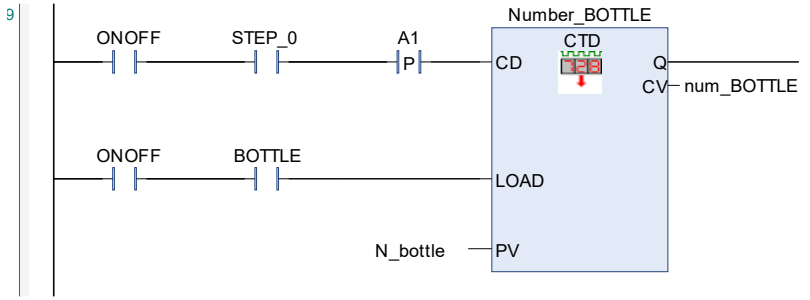
When the **stop** memory or the **machine** is switched off all the inputs of how many bottles we want to produce are set to 0



main 9 LD



This is the control for the **empty** bottles we have in input on the **first** conveyor



main

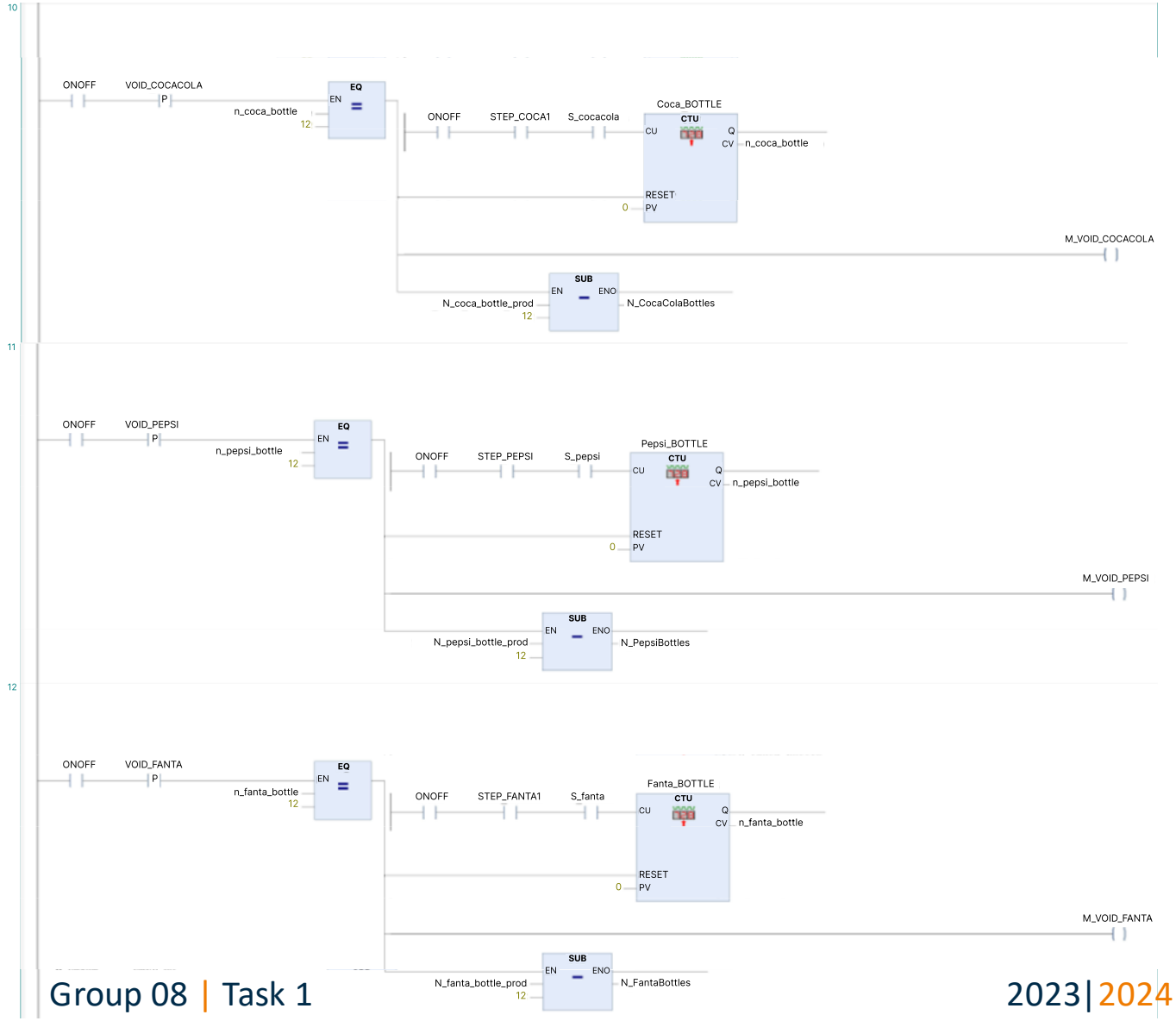
10 - 12

LD

Counters for the final buffer where the **filled** bottles are stored

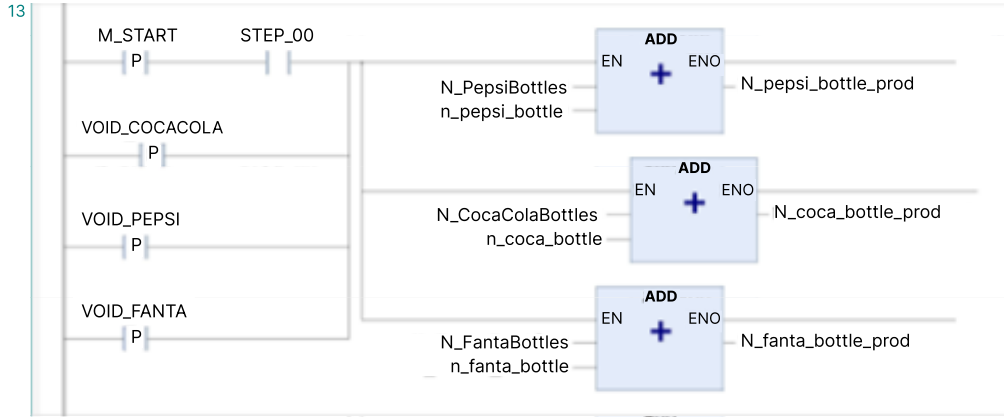
Every time a **bottle** arrives the counter is **incremented**

When the buffer is empty the value for the counter is set to zero



main 13 LD

A **variable** is updated with the number of bottle we want to have in the **buffer** when the machine will stop



Cycle management

1 - 32 ST

00

Small tank filling and waiting for the start button to be pressed; when the **start** button is pressed the big tank is filled and the step is updated to **STEP 0**

0

The bottle is **pushed** on the **conveyor**; when piston A is in position A1 the step is updated to **STEP 1**

1

When the bottle on the conveyor reaches the filling position, the **conveyor** is **stopped** and the step is updated to **STEP 2**

```
1   //  Definition  of  initial  conditions
2   IF  A0  AND  B0  AND  P0  AND  S_PEPSI  AND  NOT( EMERGENCY)  THEN
3       INITIAL_CONDITIONS  :=  TRUE;
4   ELSE
5       INITIAL_CONDITIONS  :=  FALSE ;
6   END_IF
7
8   //  definition  of  STEP_0
9   IF  ( STEP_PEPSI  OR  STEP_COCA2  OR  STEP_FANTA2 )  AND  INITIAL_CONDITIONS  THEN
10      M_STEP_00  :=  TRUE;
11  ELSE
12      M_STEP_00  :=  FALSE ;
13  END_IF
14
15  IF  STEP_00  AND  NOT( EMERGENCY)  AND  ( BIG_TANK  >  0 )  AND  M_START  AND
    ( ( ProducingCocaCola  AND  n_coca_bottle  <  12 )  OR
    ( ProducingFanta  AND  n_fanta_bottle  <  12 )  OR  ( ProducingPepsi  AND
    n_pepsi_bottle  <  12 ) )  AND  ( num_BOTTLES  >=  N_fanta_bottle_prod
    + N_coca_bottle_prod  + N_pepsi_bottle_prod  - ( n_coca_bottle
    + n_pepsi_bottle  + n_fanta_bottle  ) )  AND  ( FANTA_TANK  >=  6 )  AND  (
    PEPSI_TANK  >=  6 )  AND  ( COCACOLA_TANK >=  6 )  THEN
16      M_STEP_0  :=  TRUE;
17  ELSE
18      M_STEP_0  :=  FALSE ;
19  END_IF
20
21  IF  STEP_0  AND  NOT( EMERGENCY)  AND  A1  AND  P0  THEN
22      M_STEP_1  :=  TRUE;
23  ELSE
24      M_STEP_1  :=  FALSE ;
25  END_IF
26
27  IF  STEP_1  AND  NOT( EMERGENCY)  AND  FILLING  THEN
28      M_STEP_2  :=  TRUE;
29  ELSE
30      M_STEP_2  :=  FALSE ;
31  END_IF
32
```

Cycle management

33 - 48 ST

2 The **bottle** is **filled up**, the step updates automatically with a timer to **STEP 2.1**

2.1 After the bottle is filled up, the conveyor starts moving and the **bottle** is moved to **corking** position

3 The **conveyor** is **stopped** again and piston B starts its extension. When piston B reaches the maximum extension, the step is updated to **STEP 4**

```
33      TIMER ( IN  :=  STEP_2 ,  PT  :=  T#3S ) ;
34      M_STEP_2_1  :=  TIMER . Q;
35      ET  :=  TIMER . ET ;
36
37      IF  STEP_2_1  AND  NOT( EMERGENCY)  AND  CAPPING  THEN
38          M_STEP_3  :=  TRUE;
39      ELSE
40          M_STEP_3  :=  FALSE ;
41      END_IF
42
43      IF  STEP_3  AND  NOT( EMERGENCY)  AND  B1  THEN
44          M_STEP_4  :=  TRUE;
45      ELSE
46          M_STEP_4  :=  FALSE ;
47      END_IF
48
```

Cycle management

49 - 60 ST

4

The conveyor is started again and the bottle is pushed onto the next conveyor and the task is **updated** based on the **liquid** in the bottle

pepsi

The bottle stays in place and a person **picks** the bottle from the conveyor and puts it in a **buffer**

coca1

The conveyor is **activated** and the bottle is moved to the right position of the **buffer**. When the bottle reaches the right position the conveyor is **stopped** and the step is updated to **STEP coca2**

```
49      IF   STEP_4   AND  NOT( EMERGENCY)  AND  ProducingPepsi       AND  END  AND  S_PEPSI    THEN
50           M_STEP_PEPSI   :=   TRUE;
51      ELSE
52           M_STEP_PEPSI   :=   FALSE ;
53      END_IF
54
55      IF   STEP_4   AND  NOT( EMERGENCY)  AND  ProducingCocaCola       AND  END  AND  S_PEPSI    THEN
56           M_STEP_COCA1  :=   TRUE;
57      ELSE
58           M_STEP_COCA1  :=   FALSE ;
59      END_IF
60
```

Cycle management

61 - 78 ST

→ fanta1 →

The conveyor is **activated** and the bottle is moved to the right position of the **buffer**. When the bottle reaches the right position the conveyor is **stopped** and the step is updated to **STEP fanta2**

→ coca2 →

The bottle is **picked** from the conveyor and puts into the **buffer**. The **homing** of all the visibility pistons is performed and the step is updated to **STEP 00**

→ fanta2 →

The bottle is **picked** from the conveyor and puts into the **buffer**. The **homing** of all the visibility pistons is performed and the step is updated to **STEP 00**

```
61      IF  STEP_4  AND NOT( EMERGENCY)  AND ProducingFanta      AND END AND S_PEPSI   THEN
62          M_STEP_FANTA1  :=  TRUE;
63      ELSE
64          M_STEP_FANTA1  :=  FALSE ;
65      END_IF
66
67      IF  STEP_COCA1  AND NOT( EMERGENCY)  AND ProducingCocaCola      AND S_COCA  THEN
68          M_STEP_COCA2  :=  TRUE;
69      ELSE
70          M_STEP_COCA2  :=  FALSE ;
71      END_IF
72
73      IF  STEP_FANTA1  AND NOT( EMERGENCY)  AND ProducingFanta      AND S_FANTA  THEN
74          M_STEP_FANTA2  :=  TRUE;
75      ELSE
76          M_STEP_FANTA2  :=  FALSE ;
77      END_IF
```

Cycle management

79 - 91

ST

When the machine is switched **off** all the steps are **reset** except **STEP 00** that will be set

```
79      STEP_00    S=   ONOFF;
80      STEP_0     R=   ONOFF;
81      STEP_1     R=   ONOFF;
82      STEP_2     R=   ONOFF;
83      STEP_2_1   R=   ONOFF;
84      STEP_3     R=   ONOFF;
85      STEP_4     R=   ONOFF;
86      STEP_PEPSI    R=   ONOFF;
87      STEP_COCA1    R=   ONOFF;
88      STEP_COCA2    R=   ONOFF;
89      STEP_FANTA1   R=   ONOFF;
90      STEP_FANTA2   R=   ONOFF;
91
```

Cycle management

92 - 120

ST

The memories of the steps will **update** the step variable and the new step **resets** the previous one

```
92
93
94
95      STEP_00   S=   M_STEP_00 ;
96      STEP_0    S=   M_STEP_0 ;
97      STEP_1    S=   M_STEP_1 ;
98      STEP_2    S=   M_STEP_2 ;
99      STEP_2_1    S=   M_STEP_2_1 ;
100     STEP_3    S=   M_STEP_3 ;
101     STEP_4    S=   M_STEP_4 ;
102     STEP_PEPSI    S=   M_STEP_PEPSI ;
103     STEP_COCA1   S=   M_STEP_COCA1 ;
104     STEP_COCA2   S=   M_STEP_COCA2 ;
105     STEP_FANTA1    S=   M_STEP_FANTA1 ;
106     STEP_FANTA2    S=   M_STEP_FANTA2 ;
107
108     STEP_00   R=   STEP_0 ;
109     STEP_0    R=   STEP_1 ;
110     STEP_1    R=   STEP_2 ;
111     STEP_2    R=   STEP_2_1 ;
112     STEP_2_1    R=   STEP_3 ;
113     STEP_3    R=   STEP_4 ;
114     STEP_4    R=   ( STEP_PEPSI    OR   STEP_COCA1   OR   STEP_FANTA1 ) ;
115     STEP_PEPSI    R=   STEP_00 ;
116     STEP_COCA1   R=   STEP_COCA2 ;
117     STEP_COCA2   R=   STEP_00 ;
118     STEP_FANTA1    R=   STEP_FANTA2 ;
119     STEP_FANTA2    R=   STEP_00 ;
120
```

Actions

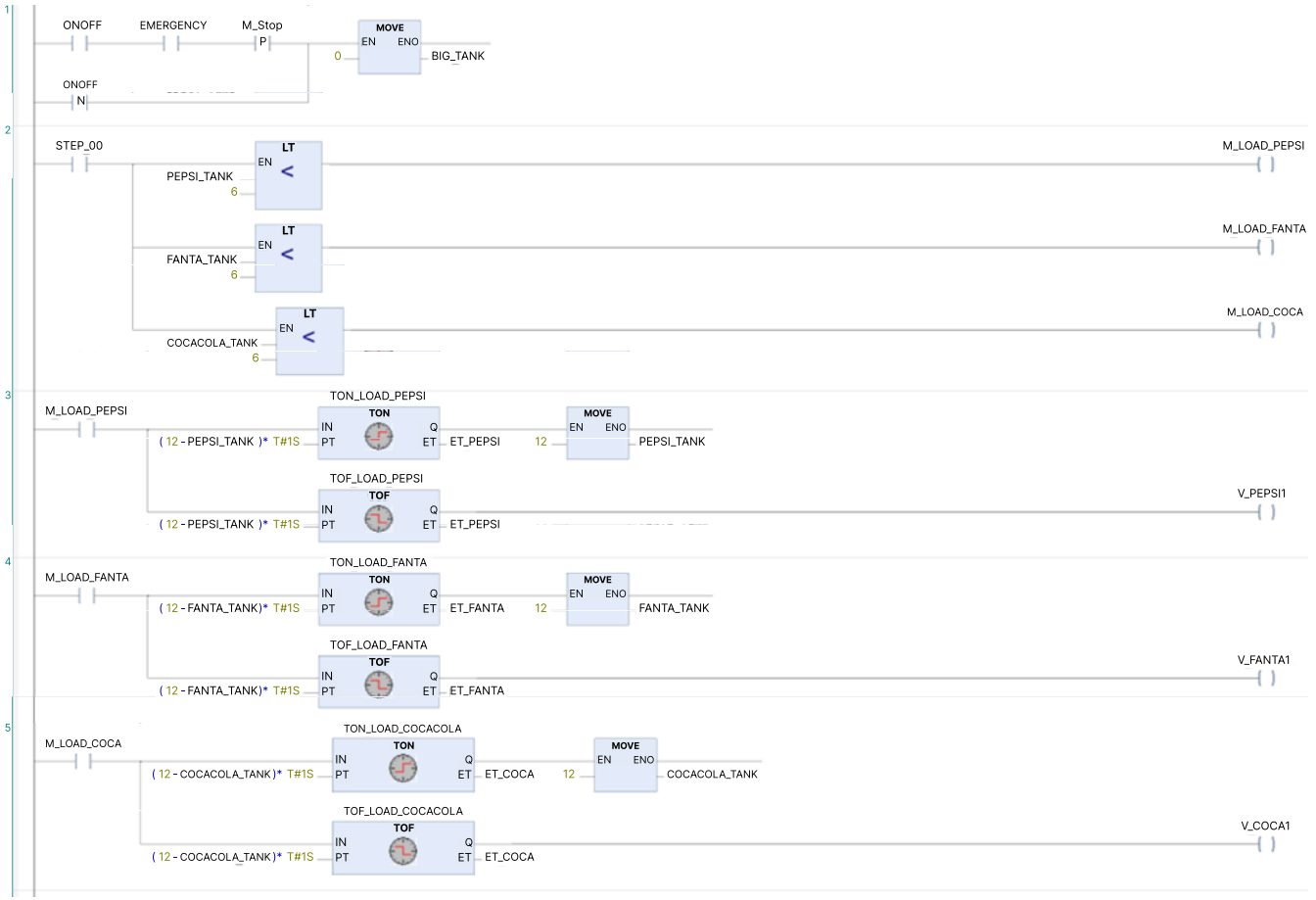
1 - 5

LD

If the tank is **full** when the machine is **stopped** the tank will be **empty**

If the tank for the cocacola, pepsi or fanta has the liquid for less than 6 bottles it is **filled again**

It is selected with which **drink** the machine will fill the bottles; if all the bottles that must be produced are produced the machine will **stop** automatically



Actions

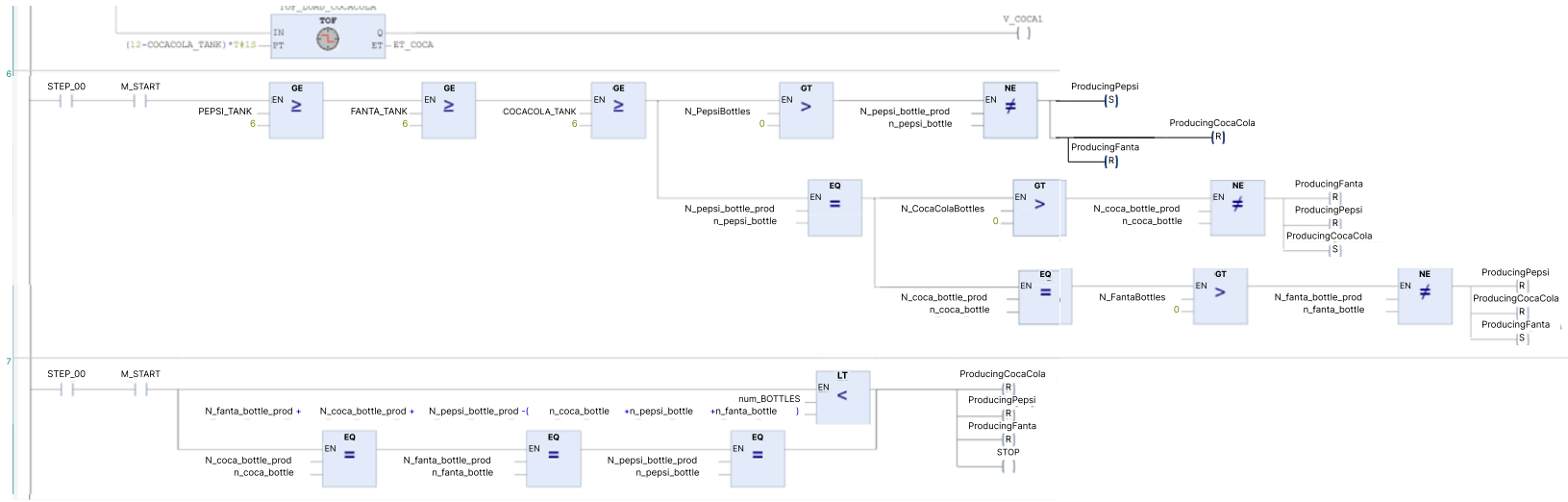
6 - 7

LD

If the tank is **full** when the machine is **stopped** the tank will be **empty**

If the tank for the cocacola, pepsi or fanta has the liquid for less than 6 bottles it is **filled again**

It is selected with which **drink** the machine will fill the bottles; if all the bottles that must be produced are produced the machine will **stop** automatically



Actions

8 - 10

LD

With the selected liquid the **tank** that will fill the bottles will be **filled** with the right amount of the liquid and the right liquid

0

Action for **STEP 0**

1

Action for **STEP 1**



Actions

11 - 13

LD

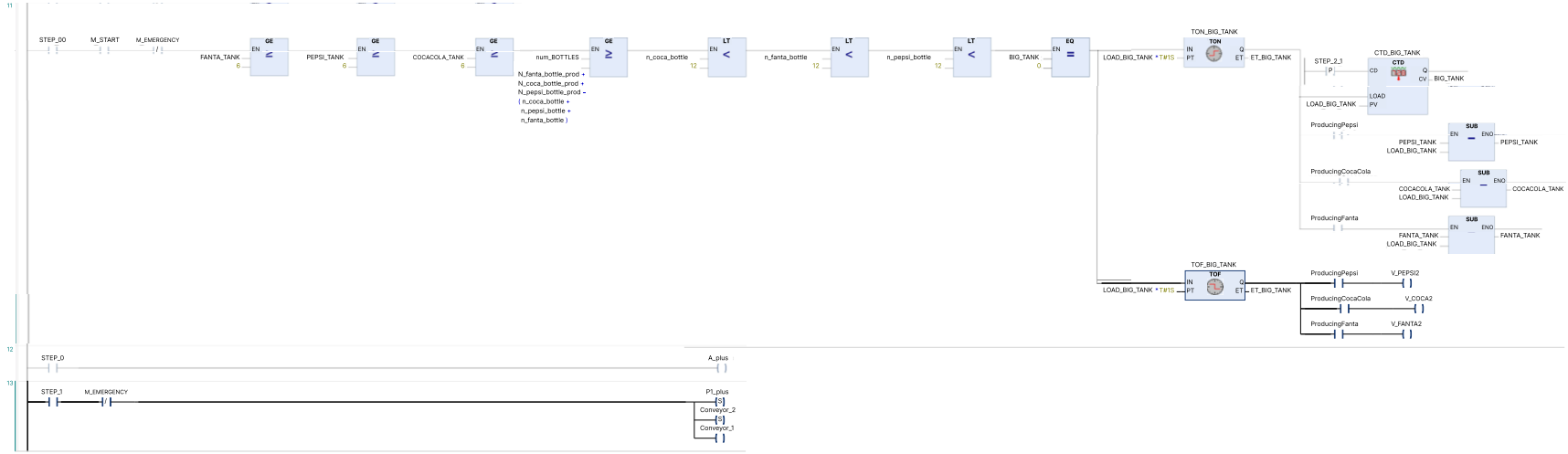
With the selected liquid the **tank** that will fill the bottles will be **filled** with the right amount of the liquid and the right liquid

0

Action for **STEP 0**

1

Action for **STEP 1**



Actions 14 - 16 LD

2 Action for **STEP 2**

2.1 Action for **STEP 2.1**

3 Action for **STEP 3**



Actions 17 - 19 LD

4 Action for **STEP 4**

coca1 Action for **STEP coca1**

coca2 Action for **STEP coca2**

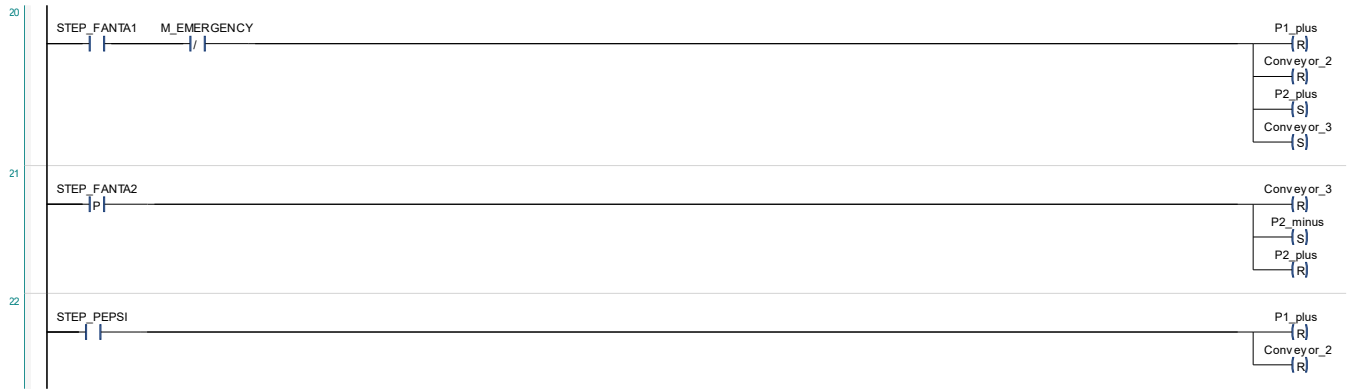


Actions 20 - 22 LD

fanta1 Action for **STEP fanta1**

fanta2 Action for **STEP fanta2**

pepsi Action for **STEP pepsi**



Actions

23 - 25

LD

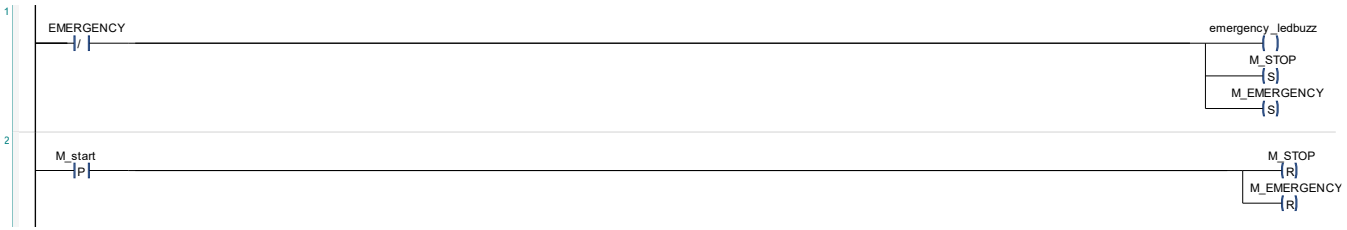
Homing of the pistons for the visibility is performed

In step 00 or when the emergency is pressed all conveyors and valves are **reset**



Emergency LD

The sequence of actions that must be performed to restart the machine after the **emergency** is shown here



After the release of the emergency button, press the start button and the cycle will **restart** from where the machine was stopped

Animation FBD

The **analog ouput** from **fluidsim** return us a value from 0 to 32761

A proportion is performed to **scale** the value

The **value** we need is the number of **pixels** that our object should move in the **visualization**



Visibility

1 - 21

ST



Performs the **switch on** of the **led** in visualization_0 that tells us if the buffer is **full**

```
1   // Logic to control object visibility based on conditions
2
3   // Visibility led clear tank
4   IF ( n_coca_bottle = 12 ) THEN
5       CLEAR_COCA_LED := TRUE ;
6   ELSE
7       CLEAR_COCA_LED := FALSE ;
8   END_IF
9
10  IF ( n_fanta_bottle = 12 ) THEN
11      CLEAR_FANTA_LED := TRUE ;
12  ELSE
13      CLEAR_FANTA_LED := FALSE ;
14  END_IF
15
16  IF ( n_pepsi_bottle = 12 ) THEN
17      CLEAR_PEPSI_LED := TRUE ;
18  ELSE
19      CLEAR_PEPSI_LED := FALSE ;
20  END_IF
21
```

Visibility

22 - 58

ST



All the variables for the visibility of the **empty bottle** on **conveyor 1** are performed

```
22      // Visibility conveyor 1
23      IF num_BOTTLE <= 9 THEN
24          BottleConveyor1A := TRUE ;
25      ELSE
26          BottleConveyor1A := FALSE ;
27      END_IF
28
29      IF num_BOTTLE <= 8 THEN
30          BottleConveyor1B := TRUE ;
31      ELSE
32          BottleConveyor1B := FALSE ;
33      END_IF
34
35      IF num_BOTTLE <= 7 THEN
36          BottleConveyor1C := TRUE ;
37      ELSE
38          BottleConveyor1C := FALSE ;
39      END_IF
40
41      IF num_BOTTLE <= 6 THEN
42          BottleConveyor1D := TRUE ;
43      ELSE
44          BottleConveyor1D := FALSE ;
45      END_IF
46
47      IF num_BOTTLE <= 5 THEN
48          BottleConveyor1E := TRUE ;
49      ELSE
50          BottleConveyor1E := FALSE ;
51      END_IF
52
53      IF num_BOTTLE <= 4 THEN
54          BottleConveyor1F := TRUE ;
55      ELSE
56          BottleConveyor1F := FALSE ;
57      END_IF
58
```

Visibility

59 - 87

ST

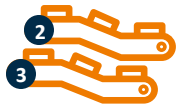


All the variables for the visibility of the **empty bottle** on **conveyor 1** are performed

```
59      IF num_BOTTLE <= 3 THEN
60          BottleConveyor1G := TRUE ;
61      ELSE
62          BottleConveyor1G := FALSE ;
63      END_IF
64
65      IF num_BOTTLE <= 2 THEN
66          BottleConveyor1H := TRUE ;
67      ELSE
68          BottleConveyor1H := FALSE ;
69      END_IF
70
71      IF num_BOTTLE <= 1 THEN
72          BottleConveyor1I := TRUE ;
73      ELSE
74          BottleConveyor1I := FALSE ;
75      END_IF
76
77      IF num_BOTTLE = 0 THEN
78          BottleConveyor1J := TRUE ;
79      ELSE
80          IF A0 OR STEP_0 THEN
81              BottleConveyor1J := FALSE ;
82          ELSE
83              BottleConveyor1J := TRUE ;
84          END_IF
85      END_IF
86
87
```

Visibility

88 - 131 ST



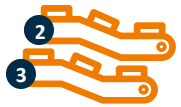
The visibility of the bottle during the cycle on **conveyor 2** and **conveyor 3** is performed



A bottle of **cocacola** has a different image from the bottle of **pepsi** or **fanta**

```
88      // Visibility Conveyor 2
89
90      IF STEP_1 OR STEP_2 THEN
91          BottleConveyor2 := FALSE ;
92      ELSE
93          BottleConveyor2 := TRUE ;
94      END_IF
95
96      IF ( STEP_2_1 OR STEP_3 ) AND ProducingCocaCola THEN
97          BottleConveyor2_CocaCola := FALSE ;
98      ELSE
99          BottleConveyor2_CocaCola := TRUE ;
100     END_IF
101
102     IF ( STEP_2_1 OR STEP_3 ) AND ProducingFanta THEN
103         BottleConveyor2_Fanta := FALSE ;
104     ELSE
105         BottleConveyor2_Fanta := TRUE ;
106     END_IF
107
108     IF ( STEP_2_1 OR STEP_3 ) AND ProducingPepsi THEN
109         BottleConveyor2_Pepsi := FALSE ;
110     ELSE
111         BottleConveyor2_Pepsi := TRUE ;
112     END_IF
113
114     IF STEP_4 AND ProducingCocaCola THEN
115         BottleCappedConveyor2_CocaCola      := FALSE ;
116     ELSE
117         BottleCappedConveyor2_CocaCola      := TRUE ;
118     END_IF
119
120     IF STEP_4 AND ProducingFanta THEN
121         BottleCappedConveyor2_Fanta := FALSE ;
122     ELSE
123         BottleCappedConveyor2_Fanta := TRUE ;
124     END_IF
125
126     IF STEP_4 AND ProducingPepsi THEN
127         BottleCappedConveyor2_Pepsi := FALSE ;
128     ELSE
129         BottleCappedConveyor2_Pepsi := TRUE ;
130     END_IF
131
```

Visibility 132 - 150 ST



The visibility of the bottle during the cycle on **conveyor 2** and **conveyor 3** is performed



A bottle of **cocacola** has a different image from the bottle of **pepsi** or **fanta**

```
132     // Conveyor 3
133     IF STEP_PEPSI THEN
134         BottleCappedConveyor3_Pepsi := FALSE ;
135     ELSE
136         BottleCappedConveyor3_Pepsi := TRUE ;
137     END_IF
138
139     IF STEP_COCA1 THEN
140         BottleCappedConveyor3_CocaCola      := FALSE ;
141     ELSE
142         BottleCappedConveyor3_CocaCola      := TRUE ;
143     END_IF
144
145     IF STEP_FANTA1 THEN
146         BottleCappedConveyor3_Fanta := FALSE ;
147     ELSE
148         BottleCappedConveyor3_Fanta := TRUE ;
149     END_IF
150
```

Visibility

151 - 187 ST

The **visualization** of the **bottles** in the buffer is performed here

The number of switched on **led** is the number of **bottles** in the **buffer**

```
151     // Buffer
152     IF n_coca_bottle >= 1 THEN
153         colaLed01 := TRUE ;
154     ELSE
155         colaLed01 := FALSE ;
156     END_IF
157
158     IF n_coca_bottle >= 2 THEN
159         colaLed02 := TRUE ;
160     ELSE
161         colaLed02 := FALSE ;
162     END_IF
163
164     IF n_coca_bottle >= 3 THEN
165         colaLed03 := TRUE ;
166     ELSE
167         colaLed03 := FALSE ;
168     END_IF
169
170     IF n_coca_bottle >= 4 THEN
171         colaLed04 := TRUE ;
172     ELSE
173         colaLed04 := FALSE ;
174     END_IF
175
176     IF n_coca_bottle >= 5 THEN
177         colaLed05 := TRUE ;
178     ELSE
179         colaLed05 := FALSE ;
180     END_IF
181
182     IF n_coca_bottle >= 6 THEN
183         colaLed06 := TRUE ;
184     ELSE
185         colaLed06 := FALSE ;
186     END_IF
187
```

Visibility

188 - 223

ST

The **visualization** of the **bottles** in the buffer is performed here



The number of switched on **led** is the number of **bottles** in the **buffer**


```
188     IF n_coca_bottle >= 7 THEN
189         colaLed07 := TRUE ;
190     ELSE
191         colaLed07 := FALSE ;
192     END_IF
193
194     IF n_coca_bottle >= 8 THEN
195         colaLed08 := TRUE ;
196     ELSE
197         colaLed08 := FALSE ;
198     END_IF
199
200     IF n_coca_bottle >= 9 THEN
201         colaLed09 := TRUE ;
202     ELSE
203         colaLed09 := FALSE ;
204     END_IF
205
206     IF n_coca_bottle >= 10 THEN
207         colaLed10 := TRUE ;
208     ELSE
209         colaLed10 := FALSE ;
210     END_IF
211
212     IF n_coca_bottle >= 11 THEN
213         colaLed11 := TRUE ;
214     ELSE
215         colaLed11 := FALSE ;
216     END_IF
217
218     IF n_coca_bottle >= 12 THEN
219         colaLed12 := TRUE ;
220     ELSE
221         colaLed12 := FALSE ;
222     END_IF
223
```

Visibility 224 - 259 ST



The **visualization** of the **bottles** in the buffer is performed here



The number of switched on **led** is the number of **bottles** in the **buffer**

```
224     IF n_pepsi_bottle >= 1 THEN
225         pepsiLed01 := TRUE ;
226     ELSE
227         pepsiLed01 := FALSE ;
228     END_IF
229
230     IF n_pepsi_bottle >= 2 THEN
231         pepsiLed02 := TRUE ;
232     ELSE
233         pepsiLed02 := FALSE ;
234     END_IF
235
236     IF n_pepsi_bottle >= 3 THEN
237         pepsiLed03 := TRUE ;
238     ELSE
239         pepsiLed03 := FALSE ;
240     END_IF
241
242     IF n_pepsi_bottle >= 4 THEN
243         pepsiLed04 := TRUE ;
244     ELSE
245         pepsiLed04 := FALSE ;
246     END_IF
247
248     IF n_pepsi_bottle >= 5 THEN
249         pepsiLed05 := TRUE ;
250     ELSE
251         pepsiLed05 := FALSE ;
252     END_IF
253
254     IF n_pepsi_bottle >= 6 THEN
255         pepsiLed06 := TRUE ;
256     ELSE
257         pepsiLed06 := FALSE ;
258     END_IF
259
```

Visibility 260 - 295 ST



The **visualization** of the **bottles** in the buffer is performed here



The number of switched on **led** is the number of **bottles** in the **buffer**

```
260     IF n_pepsi_bottle >= 7 THEN
261         pepsiLed07 := TRUE ;
262     ELSE
263         pepsiLed07 := FALSE ;
264     END_IF
265
266     IF n_pepsi_bottle >= 8 THEN
267         pepsiLed08 := TRUE ;
268     ELSE
269         pepsiLed08 := FALSE ;
270     END_IF
271
272     IF n_pepsi_bottle >= 9 THEN
273         pepsiLed09 := TRUE ;
274     ELSE
275         pepsiLed09 := FALSE ;
276     END_IF
277
278     IF n_pepsi_bottle >= 10 THEN
279         pepsiLed10 := TRUE ;
280     ELSE
281         pepsiLed10 := FALSE ;
282     END_IF
283
284     IF n_pepsi_bottle >= 11 THEN
285         pepsiLed11 := TRUE ;
286     ELSE
287         pepsiLed11 := FALSE ;
288     END_IF
289
290     IF n_pepsi_bottle >= 12 THEN
291         pepsiLed12 := TRUE ;
292     ELSE
293         pepsiLed12 := FALSE ;
294     END_IF
295
```

Visibility 296 - 331 ST



The **visualization** of the **bottles** in the buffer is performed here



The number of switched on **led** is the number of **bottles** in the **buffer**

```
296     IF n_fanta_bottle >= 1 THEN
297         fantaLed01 := TRUE ;
298     ELSE
299         fantaLed01 := FALSE ;
300     END_IF
301
302     IF n_fanta_bottle >= 2 THEN
303         fantaLed02 := TRUE ;
304     ELSE
305         fantaLed02 := FALSE ;
306     END_IF
307
308     IF n_fanta_bottle >= 3 THEN
309         fantaLed03 := TRUE ;
310     ELSE
311         fantaLed03 := FALSE ;
312     END_IF
313
314     IF n_fanta_bottle >= 4 THEN
315         fantaLed04 := TRUE ;
316     ELSE
317         fantaLed04 := FALSE ;
318     END_IF
319
320     IF n_fanta_bottle >= 5 THEN
321         fantaLed05 := TRUE ;
322     ELSE
323         fantaLed05 := FALSE ;
324     END_IF
325
326     IF n_fanta_bottle >= 6 THEN
327         fantaLed06 := TRUE ;
328     ELSE
329         fantaLed06 := FALSE ;
330     END_IF
331
```

Visibility 332 - 372 ST

The **visualization** of the **bottles** in the buffer is performed here



The number of switched on **led** is the number of **bottles** in the **buffer**


```
332     IF n_fanta_bottle >= 7 THEN
333         fantaLed07 := TRUE ;
334     ELSE
335         fantaLed07 := FALSE ;
336     END_IF
337
338     IF n_fanta_bottle >= 8 THEN
339         fantaLed08 := TRUE ;
340     ELSE
341         fantaLed08 := FALSE ;
342     END_IF
343
344     IF n_fanta_bottle >= 9 THEN
345         fantaLed09 := TRUE ;
346     ELSE
347         fantaLed09 := FALSE ;
348     END_IF
349
350     IF n_fanta_bottle >= 10 THEN
351         fantaLed10 := TRUE ;
352     ELSE
353         fantaLed10 := FALSE ;
354     END_IF
355
356     IF n_fanta_bottle >= 11 THEN
357         fantaLed11 := TRUE ;
358     ELSE
359         fantaLed11 := FALSE ;
360     END_IF
361
362     IF n_fanta_bottle >= 12 THEN
363         fantaLed12 := TRUE ;
364     ELSE
365         fantaLed12 := FALSE ;
366     END_IF
367
368
369
370
371
372
```